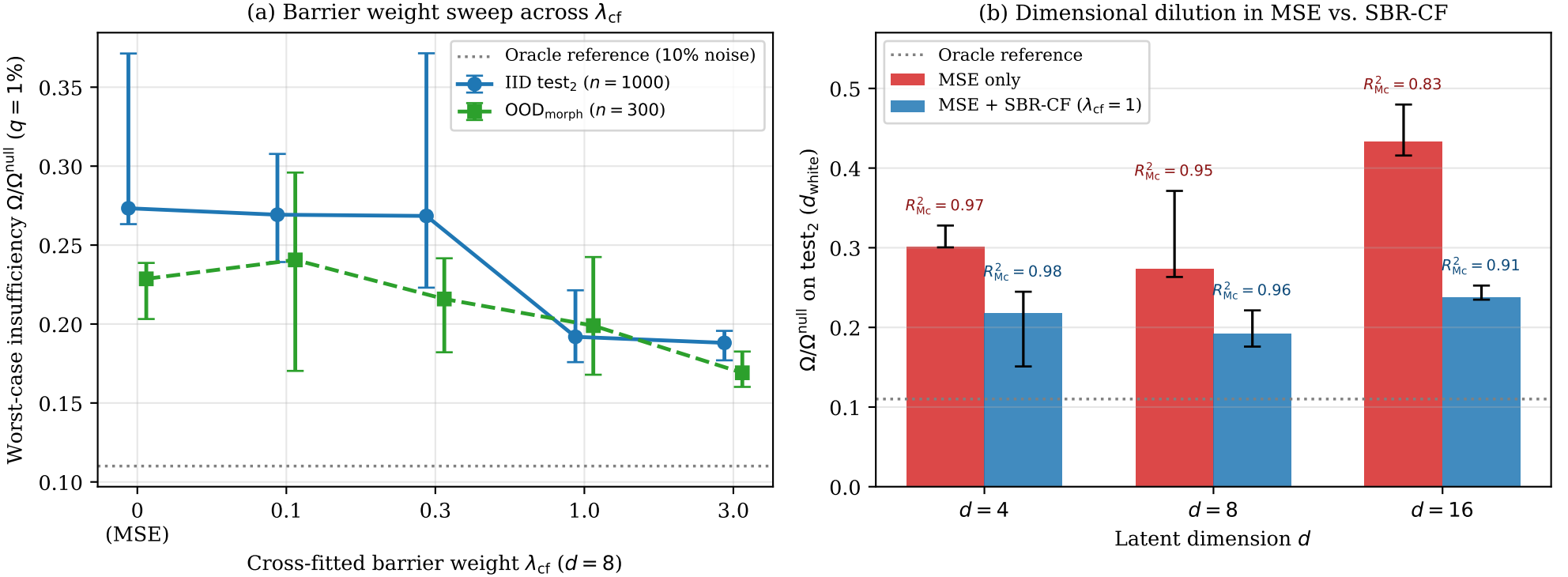
(a) Barrier weight sweep across λ_{cf}
Oracle reference (10% noise)
IID test$_2$ ($n = 1000$)
OOD$_{morph}$ ($n = 300$)
Worst-case insufficiency Ω/Ω^{null} ($q = 1\%$)
0.35
0.30
0.25
0.20
0.15
0.10
0 (MSE)
0.1
0.3
1.0
3.0
Cross-fitted barrier weight λ_{cf} ($d = 8$)
(b) Dimensional dilution in MSE vs. SBR-CF
Oracle reference
MSE only
MSE + SBR-CF ($\lambda_{cf} = 1$)
Ω/Ω^{null} on test$_2$ (d_{white})
0.5
0.4
0.3
0.2
0.1
0.0
$R^2_{Mc} = 0.97$
$R^2_{Mc} = 0.98$
$R^2_{Mc} = 0.95$
$R^2_{Mc} = 0.96$
$R^2_{Mc} = 0.83$
$R^2_{Mc} = 0.91$
$d = 4$
$d = 8$
$d = 16$
Latent dimension d